

System Design

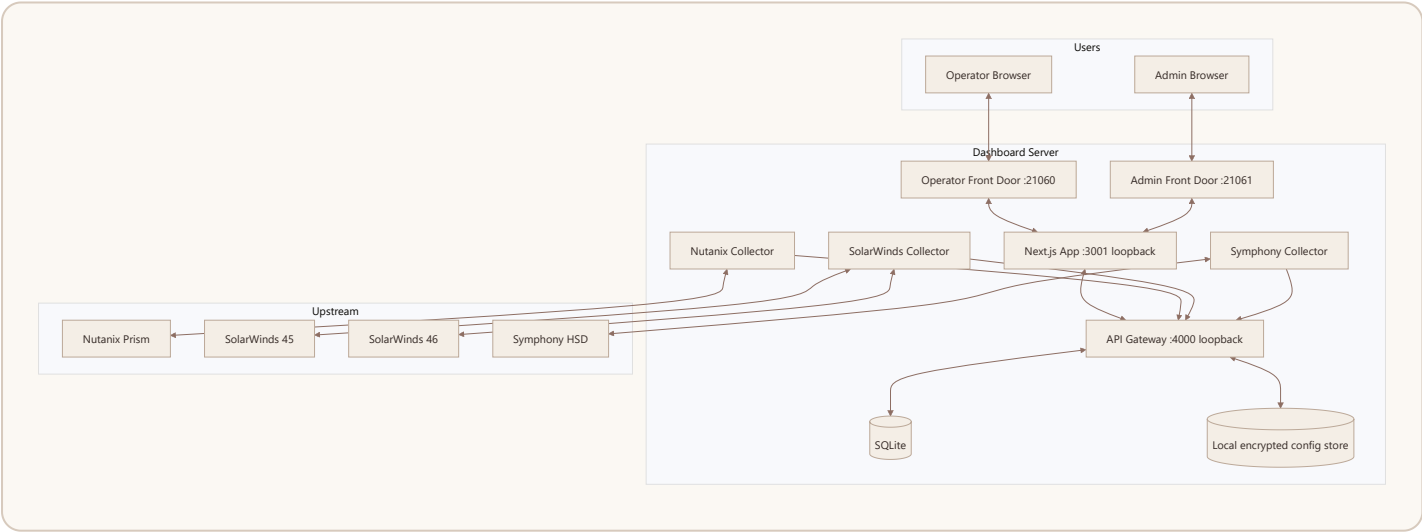
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Audience	Architects, developers, infrastructure, reviewers

1. Purpose

Describe the implemented runtime architecture, trust boundaries, data flow, deployment topology, and recovery model for the UAIL IT Dashboard.

2. System Context

The system is a multi-process Node.js application deployed on a Windows host. It presents separate operator and admin surfaces while sharing the same internal application stack.

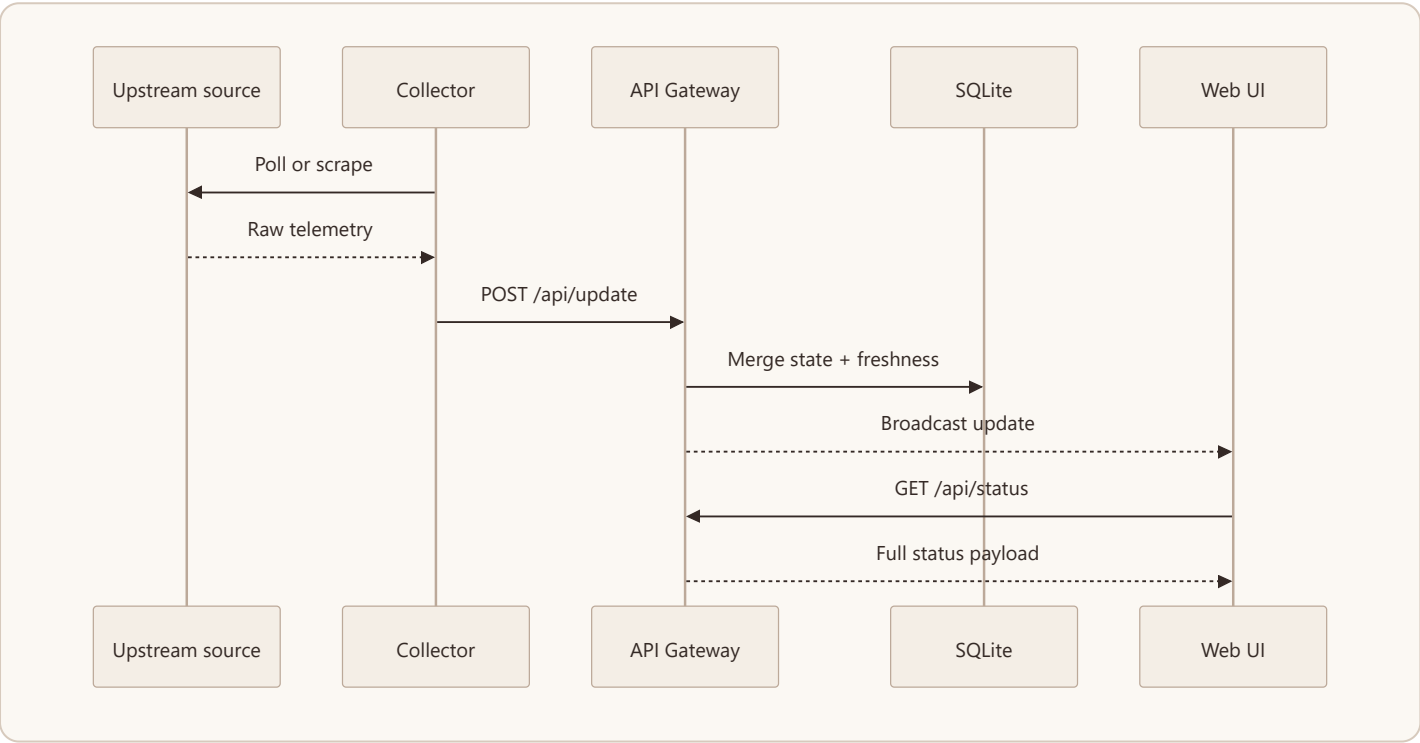


3. Runtime Components

Component	Purpose	Exposure
dashboard-frontdoor-operator	Reverse proxy and operator surface identity	LAN
dashboard-frontdoor-admin	Reverse proxy and admin surface identity	LAN

dashboard-ui	Shared Next.js web app	Loopback
api-gateway	REST API, WebSocket broadcast, admin endpoints	Loopback
nutanix-collector	Direct Nutanix API polling	Host-local
solarwinds-collector	SolarWinds portal scraping for servers and networks with independently configured portal credentials	Host-local
symphony-collector	Symphony HSD portal scraping	Host-local

4. Data Flow



5. Dashboard Domains

5.1 HCI Domain

- Source: Nutanix
- Primary content: cluster CPU, memory, storage, node state
- Trust rule: Nutanix-only

5.2 HSD Domain

- Source: Symphony HSD
- Primary content: open work items, SLA posture, special queues
- Trust rule: last-success state remains visible if collection fails

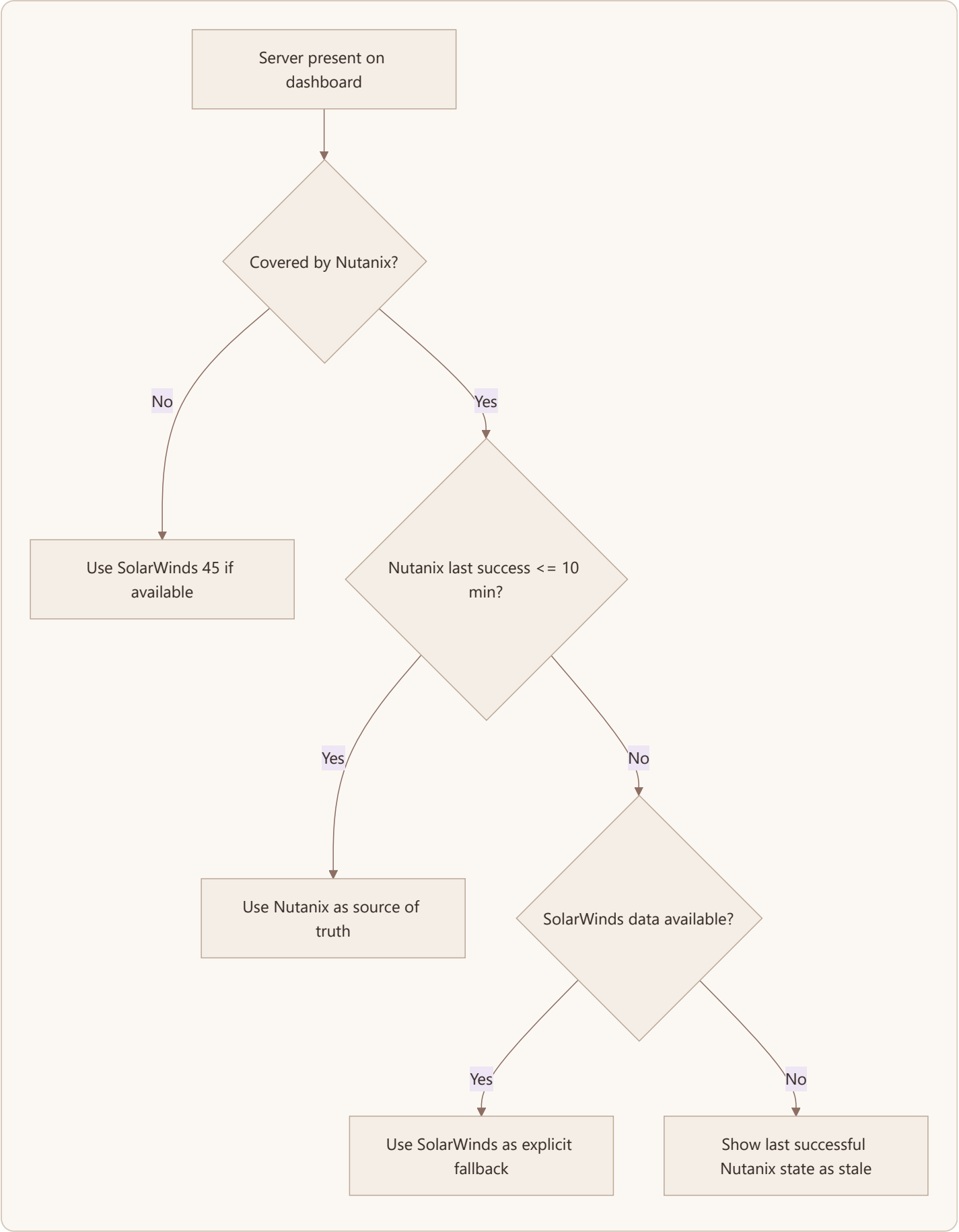
5.3 Network Domain

- Source: SolarWinds 46
- Primary content: carrier and SDWAN state, Tx/Rx, sparklines
- Trust rule: SolarWinds 46 is authoritative

5.4 Server Domain

- Primary sources:
 - Nutanix for HCI-backed servers
 - SolarWinds 45 for on-prem servers
- Fallback rule:
 - for overlapping coverage, Nutanix remains primary
 - SolarWinds is used only after Nutanix is stale for more than 10 minutes and SolarWinds data exists

6. Source Of Truth Decision Flow



7. Freshness And Failure Model

Each section is represented with operational freshness metadata:

- poll interval
- last attempt time
- last success time
- last error

The UI uses this data to derive link state and staleness messaging. The dashboard does not fabricate values; it reuses the last confirmed values and makes the freshness visible.

8. Storage Model

8.1 Primary Runtime Store

- SQLite
- current merged dashboard state
- freshness and error state

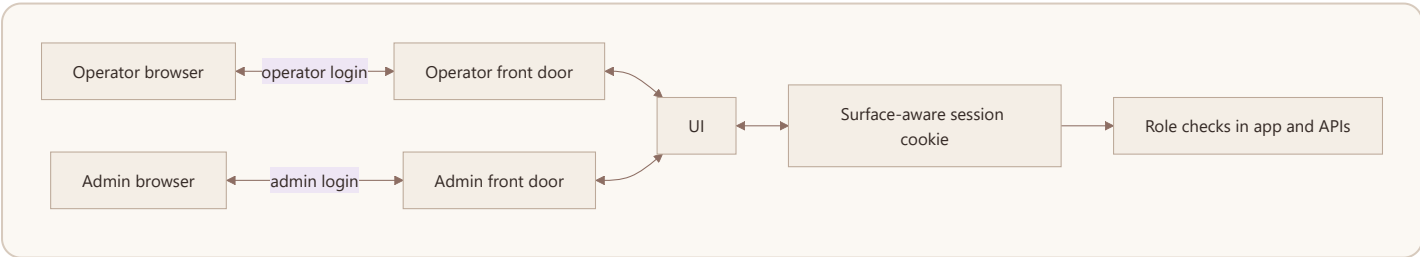
8.2 Local Config And Secrets

- runtime config and secrets stored under the shared runtime root
- encrypted local secret handling supported through the configured secret-store passphrase
- installer and bootstrap paths maintain separate credentials for Nutanix, SolarWinds 45, SolarWinds 46, and HSD

8.3 Optional Extensions

- PostgreSQL remains optional and is not required for the current production baseline

9. Authentication And Surface Separation



- operator and admin surfaces have different login prompts
- both surfaces share the same Next.js application
- the surface determines the expected role and cookie namespace
- admin APIs are restricted to admin sessions

10. Deployment Design

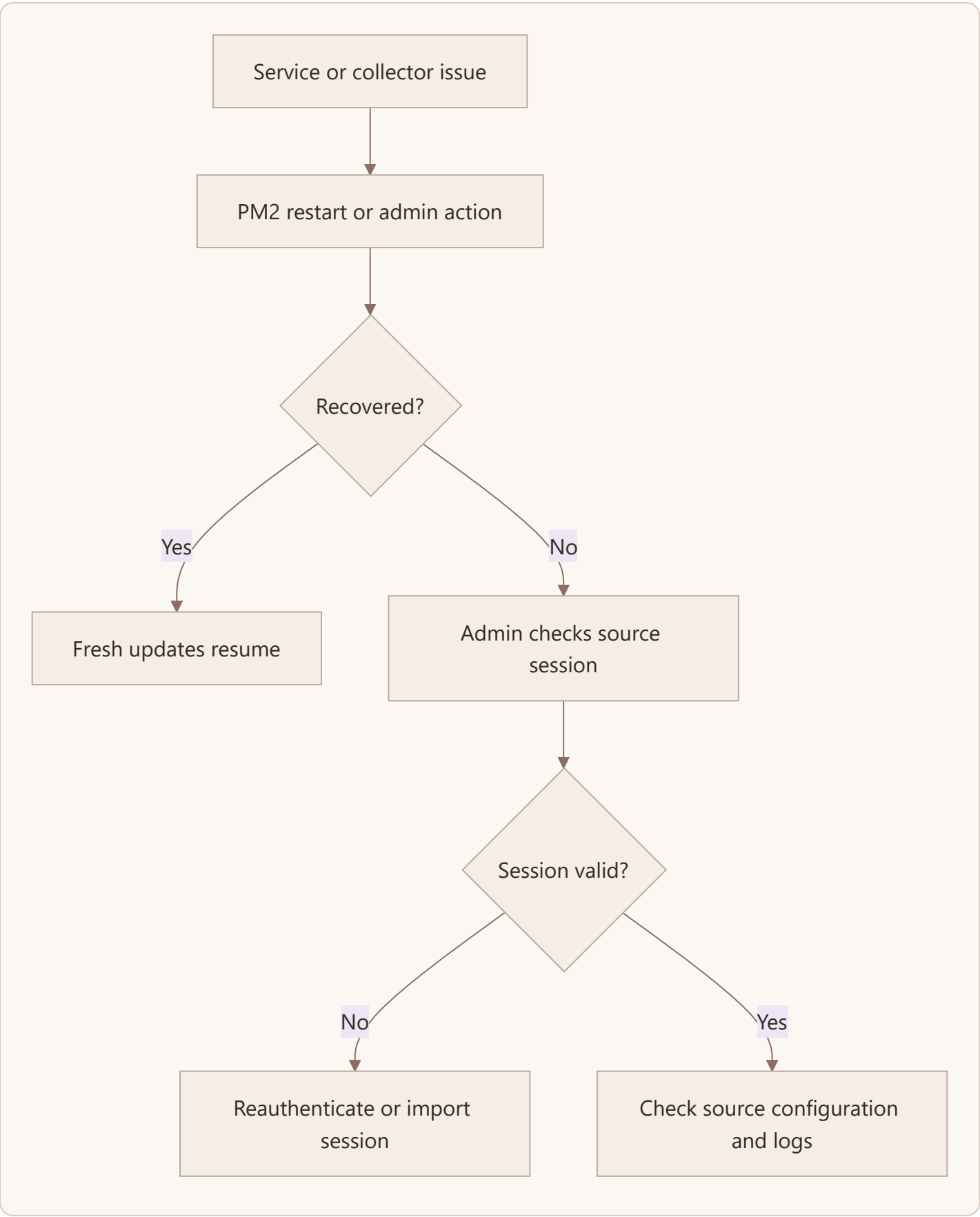
Item	Value
Primary host model	Dedicated Windows server or VM

Operator port	21060
Admin port	21061
Internal UI port	3001 loopback
Internal gateway port	4000 loopback
Process manager	PM2
Runtime store	SQLite

Bootstrap expectations:

- installer and staged deployment prompt separately for SolarWinds 45 credentials, SolarWinds 46 credentials, and HSD credentials
- generated runtime configuration uses SW_SERVERS_*, SW_NETWORKS_*, and SYM_* variables rather than assuming a shared portal identity

11. Auto-Heal And Recovery



Auto-heal mechanisms:

- PM2 process supervision
- saved process list via `pm2 save`
- runtime-user logon recovery via scheduled task and `pm2 resurrect`
- runtime permission repair for PM2 state, session files, logs, config, and app data before bootstrap
- admin-triggered restart actions

12. Security Boundaries

- only the two front-door ports should be LAN-exposed
- UI and gateway remain loopback-only
- source credentials and session state are sensitive operational assets
- the dashboard is intended for internal network deployment only

13. Key Constraints

- Nutanix TLS verification is relaxed because of upstream certificate conditions
- HSD and SolarWinds depend on browser-authenticated sessions
- current authentication is practical and local, not enterprise SSO

14. Design Principles

- show only real data from a trusted source
- make freshness explicit
- prefer operational simplicity over avoidable platform complexity
- keep admin operations available through the web surface rather than requiring shell access for normal recovery